

ARUNABHA MAJUMDER

Mechanical | Mechatronics Engineer

About Me

I design, build and test robotic hardware and exoskeletons, from the first hand sketch to a tested prototype. PhD researcher at Aalborg University.

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Aalborg, Denmark

LANGUAGE

- English (C1)
- Bengali (native)
- Hindi (spoken)

EXPERTISE

- Mechanical design and prototyping
- Stiffness analysis in SolidWorks
- Genetic algorithm optimisation
- Sensor integration, test rigs
- Motion capture and EMG testing
- Hand sketching, illustration

SOFTWARE

- SolidWorks (CSWA)
- MATLAB/Simulink, LabVIEW
- Arduino, Python (basic)
- Qualisys
- Bambu Lab, Ultimaker Cura
- Adobe Illustrator, Photoshop
- Procreate, LaTeX

EXPERIENCE

Aalborg University
PhD Researcher, Shoulder Exoskeleton
2023 - Present
Designed a hybrid exoskeleton (spring + 6 Nm motor) that uses over 25% less energy, Best Paper Award. Integrated motor, encoder and force sensors, built test rigs, and tested with 12 people. Designed and modelled a novel 5-bar planar parallel robot with VSM, estimated its stiffness with less than 10% error and quantified human arm impedance. Implemented an Arduino trigger to sync the robot with Qualisys. Taught courses and supervised students.

CSIR-CMERI, India
MTech Researcher, Pneumatic Muscles
2020 - 2022
Fabricated pneumatic muscles. Designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Siddaganga Institute of Technology
Bachelor's Project, Rehab Exoskeleton
2018 - 2019
Designed and built an upper-limb exoskeleton for a paralysed arm: CAD, machining, assembly and testing.

EDUCATION

Aalborg University, Denmark
PhD, Mechatronics and Robotics | 2023 - present

CSIR-CMERI (AcSIR), India
MTech, Mechatronics | 2020 - 2022

Siddaganga Institute of Technology, India
BTech, Mechanical Engineering | 2015 - 2019

Bangiya Sangeet Parishad, India
Senior Diploma in Fine Arts (Painting) | 2008, First Division with Distinction

AWARDS AND PUBLICATIONS

- Best Paper Award, IFToMM ISRM 2026 (Springer)
- IEEE CONECCT 2022 paper
- Best Major Project Award and TATA Technologies grant, 2019

REFERENCES

Shaoping Bai, PhD, Professor
Department of Materials and Production, Aalborg University
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Lars Rosgaard Jensen, Associate Professor
Head of Section, Physics, Mechanics and Materials; Head of Labs
Department of Materials and Production, Aalborg University